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Q&A

Breaking Boundaries: Exploring Performance Enhancement and Anti-Doping Testing in Sports

Moderator: Imir G. Metushi[a,*]; Experts: Allison Wagner,[b] Brian Ahrens,[c] Magnus Ericsson,[d] Olivier Rabin,[e] Tim Sobolevsky,[f] and Travis T. Tygart[g]

Doping in sports, a persistent challenge spanning centuries, reflects a continuous evolution in both the methods used by athletes to enhance their athletic performance and the relentless pursuit of anti-doping efforts to combat these practices. While modern awareness of doping has heightened, tracing its roots unveils a historical narrative entrenched in the pursuit of athletic excellence and the quest for an edge over competitors.

The history of doping traces back millennia, finding its origins in ancient athletic competitions like the Olympic Games, where athletes resorted to potions and tonics in the pursuit of enhanced performance. Over time, this quest has persisted, evolving into the use of modern performance-enhancing drugs, such as stimulants and anabolic steroids. Despite the lessons drawn from history and concerted efforts to educate athletes about the health risks and promote fairness, the allure of these substances endures.

At the forefront of this global battle against doping stands the World Anti-Doping Agency (WADA), established in 1999 to set and uphold standards in combating sports doping. With a mission to coordinate and promote the fight against doping, WADA accredits laboratories worldwide with presently 30 accredited labs spanning 27 countries. The anti-doping code of WADA is adopted by more than 600 sports organizations. WADA-accredited laboratories are utilized by international sports federations (e.g., Federation Internationale de Football Association), national anti-doping organizations, the International Paralympic and Olympic Committees that oversee the Paralympic/Olympic games, professional sports leagues (Major League Baseball, National Basketball Association), and other major sport event organizers that test athletes for doping. Oftentimes, athletes' accomplishments, reputations, and the virtue of fair play depend on the results of urine or blood tests. Our esteemed panel of experts unravel the complexities surrounding doping in sports, examining the historical backdrop, contemporary challenges, and concerted efforts of the scientific community to safeguard the integrity of sports through anti-doping measures.

Despite historical cases highlighting the dangers of doping and concerted international efforts promoting fair play, why do athletes continue to use performance-enhancing drugs?

Allison Wagner: Athletes face immense pressure to perform at the highest level, whether it's to win medals, secure sponsorships, receive financial gains, or maintain their professional careers. This pressure can drive some athletes, coaches, and athlete support personnel to seek any possible advantage, including turning to doping. The win-at-all cost culture is still pervasive in some sports, and athletes may also be influenced by their peers who are also doping or by a culture within their sport that tacitly condones or even encourages doping behavior. Additionally, athletes may feel pressure to "improve" their appearance with

[a]Assistant Professor, Department of Pathology and Laboratory Medicine, David Geffen School of Medicine, University of California Los Angeles, Los Angeles, CA, United States; [b]Director of Athlete and International Relations, OLY, Silver Medalist, Former World Record Holder, World Champion, NCAA Champion, United States Anti-Doping Agency, Colorado Springs, CO, United States; [c]Director of Olympic Analytical Laboratory, Department of Pathology and Laboratory Medicine, David Geffen School of Medicine, University of California Los Angeles, Los Angeles, CA, United States; [d]Director of Laboratoire AntiDopage Francais, Universite Paris-Saclay, Orsay, France; [e]Professor, Senior Director of Science and Medicine, World Anti-Doping Agency, Montreal, QC, Canada; [f]Senior Scientist at Olympic Analytical Laboratory, Department of Pathology and Laboratory Medicine, David Geffen School of Medicine, University of California Los Angeles, Los Angeles, CA, United States; [g]Chief Executive Officer, United States Anti-Doping Agency, Colorado Springs, CO, United States.
*Address correspondence to this author at: Department of Pathology and Laboratory Medicine, David Geffen School of Medicine, University of California Los Angeles, 10833 Le Conte Ave., CHS A3-231S, Los Angeles, CA 90095-1732, United States. E-mail imetushi@mednet.ucla.edu.
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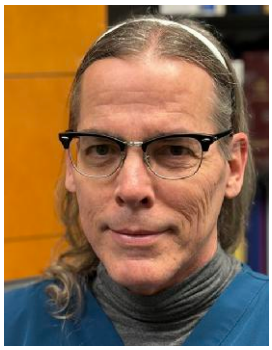
the use of prohibited performance-enhancing substances. Some athletes may be more willing to chance the health risks that often accompany the use of prohibited performance-enhancing substances and methods in favor of a possible performance benefit.



Brian Ahrens: In spite of increasing outreach and education efforts on the part of WADA and anti-doping organizations in recent years, lack of awareness regarding prohibited substances and/or the risks inherent in consumption of dietary supplements is a major contributor. For the athlete who is aware, it could be that health risks or risks to reputation seem worth taking if in their judgment all that stands between them and victory is a series of pills or an injection of a prohibited substance. The same athlete may even suspect or know that another competitor is also using and therefore feels less compelled to forswear use.

Magnus Ericcson: To effectively address the issue of athletes using performance-enhancing drugs, it's crucial to also address the cases where doping occurs from contaminated supplements or misguided advice from medical staff. To solve these problems, the most viable option is to focus more on comprehensive education of not only athletes but also all people who surround and support the athlete such as training staff, team leaders, and medical personnel.

For the real cheaters there are gains to be made. It could be financial, but I believe, most often, it is just the need for some people to win and perform rather than to make a fortune.

Olivier Rabin: There are various reasons why athletes choose to take prohibited substances. The desire to win, with all the accolades and financial benefits that it can bring, can drive athletes to consider doping. Some athletes—especially younger ones—can come under pressure from their entourage. Coaches and others can create expectations that are so great that athletes may feel that the only way to meet those standards is to take shortcuts. That is why education is so important. WADA has made values-based education of athletes and their entourage a key pillar of the fight against doping so that athletes can be provided with the right tools and positive environment to allow them to compete clean throughout their careers.

Do we have the right tests, and are they good enough to catch dopers while minimizing the chances of wrongful convictions?

Travis T. Tygart: The effectiveness of anti-doping tests is a constantly evolving effort, and significant progress has been made in recent years. Advances in detection technology have led to more sensitive and specific tests for detecting prohibited substances and methods. Techniques such as mass spectrometry and chromatography have greatly improved the ability to detect even trace amounts of banned substances in athletes' samples, and work is being done to identify where there should be threshold limits in an effort to avoid catching athletes with innocent sources of contamination. The introduction of athlete biological passport (ABP) programs has provided another tool for detecting doping. These programs monitor an athlete's biological markers over time, looking for irregularities that may indicate doping practices such as blood doping or steroid use.

Brian Ahrens: Without question the tests we are using today are the right ones for the job in the sense that they detect the overwhelming majority of performance-enhancing substances that athletes could be using and do so at very low levels. It then falls to our partners on the results management side to do their due diligence in follow-up and investigation, even including arbitration when necessary to get things right as far as any potential rules violation is concerned. That being said, room for improvement exists in terms of more economical testing or increased use of complementary alternative matrices so that it is more likely a collection occurs during the detection window of a drug and that any result is better understood.

Magnus Ericcson: For the out of competition menu, in principle, yes. We can detect many steroids for several months especially with the implementation of ABP. For the in-competition menu it is more complex,

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because of the different minimum reporting limits that exist. In my view, these quantitative aspects are more suitable for blood samples rather than urine samples. It is difficult to determine if something is taken in or out of competition based solely on urine levels, particularly when relying on data with limited administration studies. Therefore, a potential approach for the future could involve collecting dried blood spots samples in conjunction with the urine. These dried blood spots samples could then be analyzed in the event of a finding with minimum reporting limits detection in the urine, providing additional information for more precise determination.

Olivier Rabin: The anti-doping system generally works with a high degree of certainty, well beyond what is standardly applied to usual science, precisely to prevent athletes from being wrongly accused. The system is not only based on analytical science but also on results management with the right of the athlete to request reanalysis of the samples and the right to due process and a fair hearing.

Anti-doping tests and rules of application are constantly refined to ensure that optimal specificity and high sensitivity are integrated in the tests and applied by WADA-accredited laboratories.

Via its research program, WADA and its anti-doping partners constantly work to improve existing methods and develop new ones to address the constantly expanding list of substances and methods banned in sports. Since the inception of this program in 2001, WADA has dedicated more than USD 90 million to anti-doping research.

What are the main challenges the laboratory faces in effectively detecting illicit substances given the evolving landscape of new compounds and nutritional supplements?

Travis T. Tygart: There are several challenges here. Doping methods continually evolve as athletes and those involved in doping seek new substances, like designer drugs, or techniques that may not be readily detectable using current testing methods. Keeping pace with these developments requires constant research and adaptation by testing laboratories.

Nutritional supplements often contain a wide range of ingredients, including vitamins, minerals, herbs, and other compounds. Determining whether a supplement contains prohibited substances, either intentionally or inadvertently, can be challenging due to variations in manufacturing processes, labeling inaccuracies, and contamination issues.

Achieving consistency and harmonization among testing laboratories worldwide is essential for ensuring the credibility and fairness of anti-doping efforts. Addressing these challenges requires ongoing collaboration and coordination among stakeholders, including anti-doping organizations, sports federations, regulatory agencies, and testing laboratories, to develop innovative strategies and technologies for detecting illicit substances effectively while safeguarding the rights of athletes.

Brian Ahrens: One perennial challenge that comes along with new compounds is: how best to detect it? Which matrix is best suited to our needs and what metabolites might be available and then for how long? Even in a situation where a compound has been well studied in clinical trials, the analytical terrain can be very different for us. Where a pharmaceutical company will have been primarily concerned about how a drug is behaving in the human organism, our community will have to focus on "how the body is throwing the drug away" so to speak. We might even have all the answers we need to detect abuse of the drug, and then it will find its way into the nutritional supplement market either off label or as a trace contaminant and all at once we need to consider the accidental or purposeful microdosing possibility.

Tim Sobolevsky: The main challenge is the complexity of laboratory analysis dictated by today's anti-doping system. The prohibited list now includes several hundred substances and gets expanded periodically. Additionally, the analytical sensitivity requirements set by WADA get tightened. All this directly affects the cost of analysis, and the laboratory has to be capable of responding to the challenge. Ultimately, the costs add up, and it can get to a point where it is difficult for lab management to purchase new equipment and retain key scientists as resources are limited.

Having an experienced staff is essential to making the laboratory sustainable. The complex analytical methodology behind anti-doping testing requires deep knowledge to keep methods up to date. With adequate staff, the laboratory can not only satisfy current needs but also develop and maintain methods that can accommodate future challenges. Finding competent staff is itself an ongoing challenge since anti-doping analysis is a highly specialized area, and it takes years in a properly

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functioning anti-doping laboratory to perfect one's skills.

Also, the lack of reference materials for some new drugs or their metabolites can limit our ability to detect them. It takes the collective effort of the anti-doping community to source and share such materials among laboratories.

Magnus Ericcson: Funding is always of consideration. In our laboratory the challenge lies in implementing the best methods and incorporating the latest research while also persuading clients to accept the increased price for improved quality—a task that is not easy. We do have a number of test methods with very few annual requests; these require more financial support to keep running.

It often feels like our current focus is on reducing analysis costs to maintain stable prices despite inflation. Unfortunately, the funding for anti-doping testing is simply not there to order the appropriate number of tests.

Olivier Rabin: Firstly, it is not simple or straightforward to analyze several hundred substances in a small volume of a human biological sample. Secondly, current methods allow detection of compounds in the pg/mL range, which can occasionally raise the question of the origin of the substances as it could be related to the end of excretion of a doping regime or a contamination. Regarding nutritional supplements, athletes understand this class of products may present a risk of contamination. Athletes must remain cautious of this possibility when deciding to use nutritional supplements.

What key advancements or improvements are being implemented to ensure the most comprehensive and effective anti-doping measures?

Allison Wagner: Anti-doping organizations are increasingly using intelligence-based and risk assessment-driven approaches to target testing efforts more effectively. This involves gathering intelligence on doping trends, athlete behavior, and emerging threats to inform the deployment of testing resources where they are most likely to yield actionable results.

Investing in education and prevention programs is crucial for fostering a culture of clean sport and deterring athletes from doping. These programs provide athletes, coaches, and support personnel with information about the risks and consequences of doping, as well as guidance on ethical decision-making and alternatives to doping. Technological innovations, such as the development of portable testing devices and miniaturized analytical instruments, enable more flexible and efficient testing methods, including out-of-competition testing and testing in remote or challenging environments.

Collaboration among anti-doping organizations, sports federations, law enforcement agencies, and other stakeholders is essential for sharing information, intelligence, and best practices to combat doping effectively both domestically and on a global scale. Strengthening legal and ethical frameworks for anti-doping efforts is necessary to ensure transparency, fairness, and accountability in the adjudication of doping cases. This includes providing athletes with due process rights, establishing clear guidelines for sanctions and appeals, and promoting ethical standards within the sporting community.

Brian Ahrens: Increased outreach and education both locally and as part of global initiatives promises to more fully engage the athletic community in anti-doping efforts. Through involvement of athletes in the discussions regarding sample collection considerations, patterns of use and alternative testing matrices, scientists and athletes have become close strategic allies. The recent surge in utilization of capillary blood collection owes as much to its acceptance by the athletic community as it does to the pioneering research behind its use as a testing matrix. The importance of mutual understanding between the testing and athletic communities in any effort to reduce the impact of doping on sport cannot be overstated.

Tim Sobolevsky: The implementation of ABP has become a very powerful tool in the fight against doping. According to WADA, nearly 200 anti-doping rule violations were uncovered based on the ABP alone.

The discovery of long-term metabolites of common anabolic steroids, such as methandienone, stanozolol, oxandrolone, and dehydrochloromethyltestosterone, greatly improved the retrospectivity of analysis and resulted in a surge of positive cases. As a part of preventive anti-doping research, many laboratories are looking into the identification of new long-term metabolites.

Another important advancement, though not entirely scientific, is the adoption of automation in laboratories. This includes both software and hardware tools, such as sophisticated laboratory information management systems and liquid handlers or robots. Not only does this minimize the chance of human error, but it also boosts the efficiency of the laboratory and allows the personnel to spend more time on method development (which in my opinion is the best part of anti-doping science). It serves both the laboratory and its clients well; in the end, a high-throughput analysis means more tests in less time, which is particularly important for major events such as the Olympic and Paralympic Games.

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Since the majority of prohibited substances are detected using chromatography and mass spectrometry, the equipment manufacturers surely deserve credit for their efforts to improve sensitivity and speed, as high-throughput multianalyte assays would not be possible otherwise.

Olivier Rabin: WADA has made significant advancements in several of its departments over the past decade. Intelligence and investigations (I&I) is an area of growing importance for anti-doping that has delivered excellent results over the last few years. WADA has long since recognized its importance with a stand-alone and independent I&I department. The department is currently delivering strategies to increase I&I capacities and capabilities of stakeholders around the world.

Modern anti-doping is a multidisciplinary pursuit, and the various strands of the strategy are inter-woven. For example, testing and intelligence/investigations are not mutually exclusive. It is a basic principle of testing that it should be driven by intelligence, rather than being random. And many investigations involve a testing element at some point in the process. A look at the numbers of confirmed anti-doping rule violations around the world shows that the vast majority of those are cases with identification of one or more doping substances brought about by conventional in- or out-of-competition testing. In addition to the catch and sanction approach, WADA, with anti-doping partners, is putting a lot of emphasis on education of athletes with an approach based upon information and prevention of doping, with athletes remaining at the center of WADA's strategies.

How effective do you think anti-doping testing has been?

Tim Sobolevsky: The entire anti-doping system is as effective as its weakest element. Methodically, the testing itself has been fairly effective, but other areas may have been lagging behind. The recent incorporation of I&I standards into the WADA Anti-Doping Code has empowered anti-doping organizations with the ability to gather nonanalytical bits of information they can now use to strengthen their anti-doping program or as evidence to support anti-doping rule violation cases.

It is worth mentioning that different anti-doping laboratories have different analytical capabilities, even though they are well within the expectations set forth by WADA. It is not uncommon for an athlete to be tested negative in one laboratory and adverse in another. A more frequent testing of each individual athlete would help minimize those discrepancies, and this is where having high-throughput and cost-effective analytical methods is essential, as it may help anti-doping organizations collect more samples without busting their budget.

Magnus Ericcson: Significant progress has been made; however, the community can do better. The current number of tests is insufficient to effectively deter and identify all cheating athletes. Additionally, there is a disparity in the anti-doping programs globally, with some countries and sports federations investing heavily while others barely meet the minimum requirements. As a result, the quality of anti-doping efforts can vary greatly. It is imperative for WADA to intensify efforts toward harmonization in this regard.

Travis T. Tygart: The global anti-doping system has pieces that may be broken, but we would be remis not to celebrate the success of anti-doping testing. When clean athletes have their moment on the podium, the system is working. It can always be better, but a respect for the past, the big cases and small, that allow clean athletes to have trust they are competing on an even playing field is how we must measure effectiveness. Testing has been successful in detecting numerous cases of doping violations, resulting in sanctions, disqualifications, and loss of medals or records. High-profile cases serve as deterrents and reinforce the message that doping will not be tolerated in sport. In order to continue to be effective, the science has to innovate alongside the performance-enhancing techniques and drugs and must operate within ethical and legal frameworks that protect the rights of athletes while upholding the integrity of sport at all levels.

Brian Ahrens: My undoubtedly biased opinion is that anti-doping testing has been effective enough for it to remain the best opportunity we have when coupled with a strong investment in athlete education to provide an equitable and safe competitive playing field. All stakeholders must be kept in mind; there will always be room for improvement even in times of our greatest success. After being involved with testing for several years I prefer to measure its effectiveness in terms of the level of interest among athletes that it should continue.

Olivier Rabin: It is difficult to precisely assess the effectiveness of anti-doping programs. We acknowledge that much progress has been made in the scientific detection of doping in athletes' samples, in I&I that led to nonanalytical revelation of doping cases and also in education to prevent unintentional doping. The best indicator we probably have is coming from athletes themselves when they openly declare that their sport is cleaner, that now they can compete on a level playing field and win events as clean athletes.

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How does the prevalence of doping and the success of athletes using performance-enhancing substances affect the morale and mindset of clean athletes who choose not to dope, particularly when witnessing the achievements of those who do?

Allison Wagner: This can be significant. Clean athletes may experience doubt and skepticism about the legitimacy of competition and the status of a level playing field. The success of doping athletes may demotivate clean athletes, leading to feelings of disillusionment, loss of motivation to continue pursuing their athletic or life goals, depression as a result of not being able to affect systems beyond their control, and/or feeling insulted as a result of their commitment to compete fairly not being adequately protected. A sense of loss can extend beyond sport because success as an athlete requires sacrifices in all areas of life. Some athletes may grapple with ethical dilemmas related to fairness, integrity, and the values of sport. Some athletes may struggle to reconcile their commitment to competing cleanly with the allure of success and recognition achieved through doping.

The prevalence of doping can erode trust within the sporting community, as clean athletes may become distrustful of their competitors and skeptical of the integrity of sports organizations and anti-doping efforts. This erosion of trust can have long-term consequences for the credibility and legitimacy of sport.

Despite these challenges, many clean athletes remain steadfast in their commitment to competing with integrity and upholding the principles of fair play. They serve as the role models and advocates for clean sport we need them to be, inspiring others to pursue their athletic goals with honesty, integrity, and dedication.

Olivier Rabin: In past prevalence survey projects, a response often given by doped athletes is that they use prohibited substances because everyone else is and to not do so would mean losing their competitiveness. These types of WADA-led initiatives, along with the work being done by our Social Science Research team, are examples of WADA's investment in maintaining and increasing understanding of the behaviors we are seeking to reinforce or prevent. There are projects currently in the work, such as this empirical investigation of moral values and attitudes in the context of clean sport behaviors, that will provide a better understanding as to how morale and mindsets are affected.

What can the sports community do better to help promote fair play?

Travis T. Tygart: The anti-doping community must foster strong independent leadership and governance structures within sports organizations to promote transparency, accountability, and a commitment to fair play at all levels. This can be achieved by developing and enforcing ethical standards and codes of conduct that outline expected behaviors, values, and principles of fair play for athletes, coaches, officials, and administrators. No longer can we afford to let the fox guard the henhouse. We must hold individuals and organizations accountable for unethical behavior, cheating, and violations of anti-doping regulations through fair and transparent adjudication processes and appropriate sanctions and strengthen anti-doping measures by investing in comprehensive testing programs, advancing detection technologies, and implementing targeted testing strategies informed by intelligence and risk assessment.

International collaboration is essential. We must collaborate with other stakeholders, including governments, law enforcement agencies, educational institutions, and the private sector, to strengthen anti-doping efforts, share best practices, and address the root causes of doping and unethical behavior in sport, starting with the overarching governance. By taking proactive steps to promote fair play and uphold the values of integrity, respect, and honesty, the sports community can create a more inclusive, equitable, and inspiring environment for athletes and fans alike.

Olivier Rabin: WADA has placed a great emphasis on education as a key pillar of clean sport strategy as we seek to prevent doping before it starts by equipping athletes and their entourage with the tools they need to compete clean throughout their careers. Education is a core component of any anti-doping program. The introduction to the 2021 Code includes education in the prevention of intentional and unintentional doping, along with deterrence, detection, enforcement, and the rule of law. WADA has developed a number of educational initiatives to support the anti-doping community, including education policy, digital learning, and anti-doping organization capability support.

Looking ahead, how can collaborations between sports organizations, scientific communities, and regulatory bodies pave the way for more effective and comprehensive anti-doping measures?

Travis T. Tygart: Collaborations between sports organizations, scientific communities, and regulatory bodies are essential for advancing anti-doping measures and ensuring the integrity of sport.

Like the Partnership for Clean Competition, work among scientific professionals and sports leagues can facilitate research and development efforts to enhance doping detection methods, identify emerging doping trends, and develop new technologies for anti-doping testing.

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Sports organizations, scientific communities, and regulatory bodies should continue to collaborate to share information, intelligence, and best practices related to doping prevention, detection, and enforcement. Collaborative efforts can lead to the standardization of testing protocols and methodologies across different sports and jurisdictions, promoting consistency and harmonization in anti-doping efforts. This standardization can improve the reliability and credibility of testing results and facilitate mutual recognition of testing outcomes. Educational and investigative resources sharing are invaluable to the global fight for clean sport and is important for athletes' ability to trust the process and that others are being tested to the same extent as they are.

Brian Ahrens: We can expect that as advancements are made in gene therapy, the temptation to gain a competitive edge through abuse of such treatments will increase as the scale tips toward potential benefits and away from health risks. Partnerships between medical researchers and members of the anti-doping community will be vital to any efforts toward detecting abuse. Partnerships of this nature proved invaluable when new therapeutics for treatment of anemia became available and even in the case of new drug classes being studied for potential treatment of cachexia prior to having yet been deployed.

Magnus Ericcson: This is key. We need to work together for us to understand the need and also apply for funding. Regarding research, I advocate for broader engagement beyond anti-doping laboratories that can help foster innovative ideas. Additionally, for intelligence operations aimed at catching cheaters, a closer collaboration between the lab, the regulators, and the sample owner is necessary.

Olivier Rabin: A concrete example of how collaboration can pave the way for more effective anti-doping measures is the Anti-Doping I&I Project in Europe. In June 2022, WADA was awarded a grant by the European Commission to increase anti-doping I&I capacity in Europe, reduce the prevalence of doping in sport, and maximize the health benefits for European youth practicing clean sport. At its core the goal is to promote collaboration between anti-doping organizations and law enforcement agencies in Europe. This project has already delivered strong results, which were announced in March of this year. The template will then be applied in other regions of the world (Asia is next on the list).

As WADA's key activities include the harmonization of the anti-doping system, an important part of the agency's mission is to build anti-doping capacity worldwide. Not all countries have the same resources to address anti-doping—with that in mind, WADA's National and Regional Anti-Doping Program strategies are in place to establish and maintain effective relations with ADOs, implement development and capacity building programs, and support operational independence.

Along the same lines, WADA has agreements to collaborate with pharmaceutical industry partners to stay on top of the latest trends in new substances and methods being used to dope.

Is there anything else the scientific community should be doing to stop athletes from doping?

Travis Tygart: Scientists can collaborate with industry partners to develop new technologies and tools for anti-doping purposes, advocating as well for policy changes and regulatory reforms to strengthen anti-doping efforts. This may include advocating for increased funding for anti-doping research, supporting legislation to enhance anti-doping regulations, like the Rodchenkov Anti-Doping Act, and promoting international collaboration and coordination in the fight against doping.

This community has a platform to provide ethical leadership and guidance within the scientific community and sports organizations. By upholding the highest ethical standards in their research and professional practice, scientists can set a positive example for others and promote integrity in sport.

Magnus Ericcson: As scientists, it is imperative that we develop robust methods and evidence to implement these into routine testing and continuously push the boundaries of what is possible. Despite the fact that we publish our findings, making them accessible for all to benefit from—including those striving for integrity and those seeking to deceive—we must remain steadfast in our efforts. By continuing this work, we can mitigate the prevalence of doping, thereby minimizing the disparity between cheating athletes and those competing cleanly.

Olivier Rabin: As expressed in many WADA messages, anti-doping is in great need of additional resources to allow for more testing and to expand on some of the promising activities and concepts currently being developed in anti-doping. Too many athletes around the world are not tested enough in advance of or during competitions. There is a massive disconnect between the resources invested in sport and the small amount of resources dedicated to protecting clean sport.

Nonstandard Abbreviations: WADA, World Anti-Doping Agency; ABP, athlete biological passport; I&I, intelligence and investigations.

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of data; (b) drafting or revising the article for intellectual content; (c) final approval of the published article; and (d) agreement to be accountable for all aspects of the article thus ensuring that questions related to the accuracy or integrity of any part of the article are appropriately investigated and resolved. Nobody who qualifies for authorship has been omitted from the list.

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